Resampling Cannot See the User: Detecting User-Specific Hallucination in Interactive Agents with Probabilistic Commitments

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Abstract

An agent serving an individual must act on facts about that person it was never told: whether a scan is safe today, whether a sedative is needed. When it asserts such facts confidently and wrongly, nothing external can check it, because the truth lives with the user. We show that sampling-consistency detectors cannot tell users apart within a conversation: conditional on it, the model's resamples carry no information about which user it is wrong about, and for an agent that repeats its answers whether they are right or wrong, their AUROC is close to $1/2$ overall. We propose Probabilistic Game Environments (PGE), a protocol in which the agent states each assumption as an explicit commitment with a confidence and the user's implicit reactions are recorded, and an intake-aware label-event posterior that turns them into per-commitment and per-session risk. In a pre-registered evaluation on fresh sessions with five LLM agents from four families, SelfCheckGPT and semantic entropy are near chance on user-specific hallucination (0.566 [0.535, 0.595] and 0.494 [0.479, 0.510] AUROC pooled), while our posterior reaches 0.806 [0.784, 0.828] and beats the best supervised detector given the same information by +.071 [+.047, +.101]*. It also identifies *which* commitment is wrong (0.765–0.803 AUROC, calibration error 0.049–0.077) and leads at small label budgets. Exact Bayes predictors in a simulator with known truth show that the user's reactions carry information beyond the population risk of the commitments (+.087 [+.069, +.107]* AUROC), and that it is already in the conversation the agent sees: resampling fails because agreement between samples does not read it. [PENDING: clarification headline] We report every registered hypothesis, including those against our first posterior.

Keywords: LLM agents; hallucination detection; uncertainty; clarification; implicit feedback.

1 Introduction

An agent that schedules imaging appointments must decide whether a patient may have a contrast scan today. The intake form says the patient had a recent treatment, but not when. A careful agent would say it does not know; many instead answer "safe, $p = 0.85$" and proceed. We call this a *user-specific hallucination*: a confident assertion about the current user whose truth depends on something the user never disclosed. It differs from the factual hallucinations that dominate the literature (Ji et al., 2023): there is no document or search engine to check it against.

The leading black-box detectors resample the model and flag answers it does not reproduce (Manakul et al., 2023; Kuhn et al., 2023; Farquhar et al., 2024). That presumes hallucinations are stochastic. Fabricated user facts need not be: an underdetermined question maps the same context to the same guess every time. More fundamentally, two users who share a conversation (the same intake answers and, where the agent is shown them, the same reactions) are indistinguishable to the model, so nothing computed from the model and its context can tell which of them it is wrong about (Theorem 1). Context can still reveal *population* risk, for instance that "no sedation needed" is wrong for a third of patients, and we show that a well-calibrated context-only predictor can exploit it. Sampling-consistency scores of a self-consistent agent do not even do that (Corollary 1).

The way out is to look at the user rather than the model. In PGE the agent must externalize each assumption as a *commitment* $(\text{attribute}, \text{value}, p)$, and the user's reaction (accepting it, contradicting it, or ignoring it) is recorded. Reactions are noisy and partial, but they depend on the truth for *this* user, which no resample does: under a simple model the posterior of the hallucination event uses them exactly (Proposition 1), and less noisy reactions cannot hurt (Theorem 2). Our contributions:

1. **The failure, proved and measured.** No detector computed from the model and its context (resampling, self-verification, hidden-state probes) can tell which user it is wrong about within a conversation (Theorem 1), and for a self-consistent agent sampling scores are near chance overall (Corollary 1). On fresh sessions from four agents, all three resamples agree on 50–86% of the commitments whose truth depends on undisclosed facts, resample disagreement predicts their errors at 0.517–0.555 AUROC, and SelfCheckGPT and semantic entropy score 0.566 [0.535, 0.595] and 0.494 [0.479, 0.510] on user-specific hallucination.

2. **A protocol that makes these errors detectable.** In PGE the agent states each assumption as a commitment with a confidence and the user's implicit reac-

tion is recorded. Detectors on these observables beat both sampling detectors by +.090 [+.056, +.123]* to +.211 [+.179, +.242]* AUROC pooled, and the user's reactions add signal (+.072 [+.054, +.090]*; significant for 4 of 4 agents).

3. **A posterior that wins, and says which claim is wrong.** Our label-event posterior is exact under a conditional-independence model and identifies its priors from reactions alone (Proposition 2). Its intake-aware form, V7, learns which commitments the intake leaves to the user. Registered before use, it reaches 0.806 [0.784, 0.828] AUROC on user-specific hallucination, beats the strongest supervised detector given the same information by +.071 [+.047, +.101]* and SelfCheckGPT by +.240 [+.206, +.276]*, leads with 25 labelled sessions, and per commitment reaches 0.765–0.803 AUROC with calibration error 0.049–0.077. Its risk chooses what to ask the user, which we measure with the agent re-reasoning after each answer.

4. **Where the information is.** Exact Bayes predictors in a simulator with known truth show that the reactions carry information beyond the population risk of the commitments (+.087 [+.069, +.107]* AUROC), and that it is already in the conversation the agent sees: sampling fails not for lack of information, but because agreement between resamples does not read it.

5. **A pre-registered evaluation, fully reported.** After a blind internal audit of our development results we fixed a protocol bug, registered every comparison and its wording, and generated fresh sessions with five agents from four families and two domains. We report every registered outcome, including those against our posterior and two simulated patients that failed their quality gate.

2 Related Work

Hallucination detection. Sampling-based detectors measure agreement across resamples (Manakul et al., 2023; Kuhn et al., 2023; Farquhar et al., 2024); semantic entropy is explicitly scoped to stochastic confabulations, not consistent errors (Farquhar et al., 2024). Learned confidence models from a model's generations (Ulmer et al., 2024), white-box probes (Azaria and Mitchell, 2023; Chen et al., 2024; Kossen et al., 2024), and verification by the model's own questions (Dhuliawala et al., 2024) or by search (Wei et al., 2024) are all functions of the model given its context, so Theorem 1 applies to them. Consistent errors are documented in snowballing (Zhang et al., 2024) and self-contradiction studies (Mündler et al., 2024), and calibrated models must hallucinate on facts absent from training (Kalai and Vempala, 2024); our setting is the per-user analogue. Verbalized confidence (Kadavath et al., 2022; Tian et al., 2023; Xiong et al., 2024) is a commitment-level signal we include as a baseline, and iterative prompting can separate epistemic from aleatoric uncertainty (Abbasi Yadkori et al., 2024), still from the model given its context.

Hallucination in agents. MIRAGE-Bench (Zhang et al., 2025) taxonomizes agent actions unfaithful to instructions, history or observations; CAR-bench (Kirmayr et al., 2026) tests whether agents admit missing information with a simulated user; AgentHallu (Liu et al., 2026) localizes the step that introduced an error. In multiagent systems a commitment is a public, checkable obligation of one agent to another (Singh, 1999); ours are probabilistic claims about the counterpart, checked by its reactions. We study detection when the missing information is the user's own.

Clarification. When to ask has been framed through expected value of perfect information (Rao and Daumé III, 2018; Suri et al., 2025), ambiguity detection (Zhang and Choi, 2025), conformal prediction (Ren et al., 2023), and information-gain rewards on τ-bench (Deng et al., 2026; Yao et al., 2024). In clinical dialogue, MediQ (Li et al., 2024) and Uncertainty of Thoughts (Hu et al., 2024) ask a simulated patient questions to reduce the doctor model's uncertainty. These reason about the agent's uncertainty before it acts; we use post-hoc commitment risk to choose which assertions to confirm, and measure the effect with the agent re-reasoning.

Implicit and noisy feedback. Implicit signals such as clicks are informative but biased (Joachims et al., 2005). Spoken dialogue systems confirm hypotheses implicitly and update beliefs from the user's reaction (Bohus and Rudnicky, 2005; Williams and Young, 2007); our reactions play the same role for asserted user facts. The general channel of our posterior, with error-dependent contradiction and confirmation rates, has the form of the guess and slip parameters of knowledge tracing (Corbett and Anderson, 1994). Our estimation step inverts a noise channel, as in learning from noisy labels (Angluin and Laird, 1988; Natarajan et al., 2013; Scott et al., 2013) and annotator models (Dawid and Skene, 1979). Agents that revise under user pushback raise sycophancy concerns (Sharma et al., 2024); our simulated user follows a long line of user simulation for dialogue (Schatzmann et al., 2006).

3 Problem, Protocol, and Labels

Definition 1 (User-specific hallucination)**.** *A* commitment *asserts a value v for an attribute a of the current user with confidence* $p \in [0,1]$*. It is* user-dependent *if its true value is not determined by what the user has disclosed, and it is a user-specific hallucination if it is user-dependent, confident (*$p \geq \tau$*), and wrong.*

Protocol. A PGE session has four stages. (i) *Intake*: the user answers a few coarse questions; the agent sees only these answers. (ii) *Commitment*: at each turn the agent emits one commitment in a fixed format, `[attribute]: value, p=confidence`. (iii) *Reaction*: the user's implicit reaction $F_t \in \{-1, 0, +1\}$ (contradicts, no reaction, accepts) is recorded and shown to the agent for its next six turns. (iv) *Detection and intervention*: a detector scores the session and its commitments; high-risk commitments trigger a confirmation question.

Labels. With confident set $H = \{t : p_t \geq \tau\}$, $\tau = 0.6$, the *composite* label L (Eq. (1)) marks a session in which at least half of the confident commitments are wrong, and is 0 if there are none. It mixes two kinds of error. Some attributes are determined by the intake through rules the agent was never shown (e.g. contrast is needed iff urgency is medium or high); errors there are guesses about the clinic, not about the user. The *user-specific* label L_U applies the same rule to the confident commitments that are user-dependent in that session, and is 0 when there are none. Whether a commitment is user-dependent is decided per session by enumerating the hidden profile's support: `scan_safe_now`, for example, depends on days since treatment only when a treatment was recent. L_U is our primary label; L is reported as secondary.

4 What Resampling and Feedback Can Reveal

Setup. A user u is drawn from a population. At turn t the agent conditions on a context X_t: the coarse intake $\phi(u)$, the question, and the interaction history, which in PGE displays earlier reactions; X denotes $(X_1, \ldots, X_T)$. It emits a commitment $c_t = (a_t, v_t, p_t)$ (attribute, value, reported confidence) from its policy $\pi(\cdot \mid X_t)$. Apart from the truth and the reactions, nothing depends on u except through X. The truth $y_a(u)$ is private to the user; the commitment is wrong if $w_t = \mathbf{1}[v_t \neq y_{a_t}(u)]$. With confident set $H = \{t : p_t \geq \tau\}$ and error fraction ρ, a session hallucinates iff

$$L = \mathbf{1}\Big[\,|H| \geq 1 \text{ and } \textstyle\sum_{t\in H} w_t \;\geq\; \lceil \rho\,|H| \rceil\,\Big]. \qquad (1)$$

The results below hold for any binary label determined by (C, u): a single w_t, or L_A, Eq. (1) with H replaced by $H_A = \{t \in H : a_t \in A\}$ for an attribute set A determined by X (for the user-specific label L_U, the attributes the intake does not determine). We write L.

A *context-and-sample detector* is any $S = h(X, C, Z, \xi)$ with ξ independent randomness and Z drawn, given (u, X, C), from a kernel $K(\cdot \mid X, C)$ that does not involve u: resamples at any temperature, token log-probabilities, hidden states. A *feedback detector* also sees reactions $F_t \in \{-1, 0, +1\}$ (contradicts, none, accepts). By the Neyman–Pearson (NP) lemma (Neyman and Pearson, 1933), among all, possibly randomized, detectors that are functions of an observation O, thresholding $P(L = 1 \mid O)$ (randomized at ties) has the highest true-positive rate at every false-positive rate, hence the highest AUROC; ties count $\frac{1}{2}$. Theorem 1 is the standard data-processing consequence of NP (Neyman and Pearson, 1933; Blackwell, 1953), stated for this setting.

Theorem 1 (Self-consistency bound)**.** *Let $\eta_\varnothing = P(L = 1 \mid X, C)$ and S be a context-and-sample detector. Given (X, C), S is independent of u and L, so within a context it ranks hallucinating above clean sessions with probability exactly $\frac{1}{2}$. Across contexts, its ROC curve lies pointwise below that of the Bayes predictor $\eta_\varnothing$.*

Proof. Given (u, X, C), S depends only on (Z, ξ), whose law does not involve u, and L is a function of (C, u). So the law of S given (X, C) does not depend on L, $P(L = 1 \mid X, C, S) = \eta_\varnothing$, and NP applies to $O = (X, C, S)$. □

Since X contains the reactions the agent was shown, $\eta_\varnothing$ is not feedback-free, and detectors built from (X, C), such as context-only classifiers, can reflect them and exploit base rates across contexts. Resampling cannot add to (X, C) or separate users who share a context; a feedback detector reads the reactions directly, including the last. For dispersion scores, more is true:

Corollary 1 (Dispersion under systematic fabrication)**.** *Let S be a score of the commitments and their k resamples (agreement rate, semantic entropy, SelfCheck-style inconsistency) that takes a fixed value $s^\star$ whenever every resample equals its commitment, and $e_\ell = P(S \neq s^\star \mid L = \ell)$. Then*

$$\big|\mathrm{AUROC}(S) - \tfrac{1}{2}\big| \;\leq\; \tfrac{1}{2}\big(1 - (1-e_0)(1-e_1)\big) \;\leq\; \tfrac{1}{2}(e_0 + e_1).$$

If $s^\star$ is the least suspicious value of S, then $\big|\mathrm{AUROC}(S) - \frac{1}{2} - \frac{1}{2}(e_1 - e_0)\big| \leq \frac{1}{2}e_0e_1$. If every turn's resampling distribution is a point mass at the committed value, $\mathrm{AUROC}(S) = \frac{1}{2}$ however often the commitments are wrong.

Proof. For independent positive and negative draws, both scores equal $s^\star$ with probability $(1-e_0)(1-e_1)$; such pairs are ties and contribute exactly $\frac{1}{2}$. All other pairs contribute a value in $[0, 1]$. If $s^\star$ is least suspicious, a pair where only the positive (negative) differs from $s^\star$ contributes 1 (0); only pairs where both differ, with probability e_0e_1, are free. □

Table 1 checks the corollary on every agent.

Theorem 2 is Blackwell's garbling theorem (Blackwell, 1953) for a symmetric channel. Let F^ε be feedback generated for the detector: given (u, X, C), which F_t^ε are 0 has a law depending only on (X, C), and each other F_t^ε is $1 - 2w_t$ with its sign flipped independently with probability ε. Let $\eta_\varepsilon = P(L = 1 \mid X, C, F^\varepsilon)$ ($\varepsilon = 0$ is perfect feedback).

Theorem 2 (Less noisy feedback cannot hurt)**.** *Fix the joint law of (u, X, C); in particular, it does not depend on ε. For $0 \leq \varepsilon \leq \varepsilon' \leq \frac{1}{2}$ the ROC curve of η_ε lies pointwise above that of $\eta_{\varepsilon'}$, which lies above that of $\eta_\varnothing$, and $\eta_{1/2} = \eta_\varnothing$ almost surely.*

Proof. If $\varepsilon' = \varepsilon$ there is nothing to show. Otherwise $\varepsilon < \frac{1}{2}$, and flipping each nonzero F_t^{ε} independently with probability $\delta = (\varepsilon' - \varepsilon)/(1-2\varepsilon) \in [0, \frac{1}{2}]$ yields flip rate $\varepsilon + \delta(1-2\varepsilon) = \varepsilon'$, hence the joint law of $(u, X, C, F^{\varepsilon'})$. So every detector using $F^{\varepsilon'}$ is a randomized detector using F^{ε}, and NP gives the first inequality; ignoring F gives the second. At $\varepsilon = \frac{1}{2}$, F^{ε} is independent of u given (X, C): whether a reaction occurs depends only on (X, C), and every observed sign is a fair coin. □

The premise is essential: if the agent conditions on the same reactions, the law of (X, C) changes with ε and monotonicity can fail. In a two-turn example where the agent copies the reaction to an unconfident answer into a confident one on a correlated attribute without feedback, AUROC(η_ε) is 0.50 at $\varepsilon = 0$ and 0.71 at $\varepsilon = 0.25$. In PGE, Theorem 2 thus covers regenerating feedback for the detector with the agent's trajectories fixed.

The label-event posterior. Let $\kappa_t = (a_t, v_t, b(p_t))$ be the detector-visible *cell*, where $b(\cdot)$ buckets confidence, and $\mathscr{H}_t = (X_t, c_{1:t}, F_{<t}, w_{<t})$ the history up to the t-th commitment with the earlier errors. (A1) Given $\mathscr{H}_t$, $w_t \sim$ Bernoulli(π_{κ_t}). (A2) Given $(\mathscr{H}_t, w_t)$, $F_t = 0$ with a probability that depends only on a_t, and otherwise $F_t = 1 - 2w_t$ with its sign flipped with probability ε. Up to a factor free of w_t, the likelihood of F_t is $\ell_{w_t}(F_t)$ with $\ell_w(0) = 1$, $\ell_1(-1) = \ell_0(+1) = 1 - \varepsilon$ and $\ell_1(+1) = \ell_0(-1) = \varepsilon$. (A1) excludes error cascades and intake information beyond the cell, but lets the agent choose cells after seeing feedback.

Proposition 1 (Exact posterior)**.** *Under (A1)–(A2), $P(w_{1:T} \mid X, C, F) = \prod_t q_t^{w_t}(1-q_t)^{1-w_t}$ with $q_t = \pi_{\kappa_t}\ell_1(F_t)/\big(\pi_{\kappa_t}\ell_1(F_t) + (1-\pi_{\kappa_t})\ell_0(F_t)\big)$. Hence $\eta_\varepsilon = P(L = 1 \mid X, C, F)$ is 0 if $H = \varnothing$ and otherwise the Poisson-binomial tail $P\big(\sum_{t\in H} B_t \geq \lceil \rho|H| \rceil\big)$, $B_t \sim$ Bernoulli(q_t) independent, computable in $O(|H|^2)$. It is Bayes-optimal under (A1)–(A2) with the true priors and channel; the implementation plugs in estimates.*

Proof. Write the joint law in the order $c_1, w_1, F_1, c_2, \ldots$. The policy and the questions see u only through X, so $p(X, C, F, w) = A(X, C, F)\prod_t \pi_{\kappa_t}^{w_t}(1-\pi_{\kappa_t})^{1-w_t}\ell_{w_t}(F_t)$, where A collects the intake prior, the questions, the policy and the w-free factors of (A2). Normalizing over $w \in \{0,1\}^T$ gives independent Bernoulli(q_t). H is fixed by C, L depends on w only through $\sum_{t\in H} w_t$, and NP gives optimality. □

The same proof gives the posterior of L_A (the tail over H_A), and holds for any channel $P(F_t \mid \mathscr{H}_t, w_t) = \ell_{w_t}(F_t)$ that may depend on a_t, as in the general-channel detector. Under (A1) this user's reactions enter only through $\ell_1(F_t)/\ell_0(F_t)$; setting these to 1 leaves the population base rates π_κ (the prior-only ablation). Cascades violate (A1), but a simple cascade model does not help (§7.7).

Proposition 2 (Priors are identified where feedback exists)**.** *Assume (A2) and let $\pi_\kappa = P(w_t = 1 \mid \kappa_t = \kappa)$. Let n_κ count the commitments in cell κ that received a reaction ($F_t \neq 0$) and $\hat{r}_\kappa$ the fraction contradicted, and assume their errors are independent given n_κ (implied by (A1), or by i.i.d. sessions in which each attribute is committed at most once). Then the contradiction rate is $r_\kappa = \varepsilon + (1-2\varepsilon)\pi_\kappa$, and for known $\varepsilon < \frac{1}{2}$, $\hat{\pi}_\kappa = (\hat{r}_\kappa - \varepsilon)/(1-2\varepsilon)$ needs no labels and, given n_κ, with probability at least $1-\delta$ (Hoeffding, 1963),*

$$|\hat{\pi}_\kappa - \pi_\kappa| \;\leq\; \frac{1}{1-2\varepsilon}\sqrt{\frac{\log(2/\delta)}{2n_\kappa}};$$

projecting onto $[0,1]$ does not increase the error. Cells on attributes that never receive feedback are not identified. If ε is unknown, the reactions depend on (π, ε) only through $(r_\kappa)_\kappa$, so every $\varepsilon' \in [0, \frac{1}{2})$ with $\varepsilon' \leq \min_\kappa \min(r_\kappa, 1-r_\kappa)$, with $\pi'_\kappa = (r_\kappa - \varepsilon')/(1-2\varepsilon')$, fits equally well.

One scalar anchor (ε, or one cell's $\pi_\kappa \neq \frac{1}{2}$) resolves this; we choose ε by training-split label likelihood. Unidentified cells get the bucket prior pooled over the other attributes. The bound covers the raw estimate, not the implementation's clipping to $[0.01, 0.99]$ and shrinkage toward coarser cells (value → attribute → bucket).

5 Detectors

PGE yields two observables per turn, a commitment and a reaction. We read them with two kinds of *PGE detectors*: a label-event posterior, which also gives per-commitment risk, and supervised classifiers. The baselines read only the model.

PGE posterior. The score is a plug-in estimate of the posterior of Proposition 1. Each confident commitment is assigned to a cell $\kappa = (\text{attribute}, \text{value}, \text{confidence bucket})$ with buckets $[0.6, 0.65)$, $[0.65, 0.75)$, $[0.75, 1]$. Cell priors are estimated from training reactions by the method of moments (Proposition 2), shrunk toward attribute-level and bucket-level estimates (strength $k = 5$); cells on the two attributes that never receive reactions are not identified and get the pooled bucket-level prior. Our pre-registered primary detector, V6, uses a general channel: the contradiction rates $r_w = P(F = -1 \mid w)$ are chosen on training labels over a grid ($r_1 > r_0 + 0.1$), priors are $\pi = (\text{rate} - r_0)/(r_1 - r_0)$, and the confirmation rates $g_w = P(F = +1 \mid w)$ are fit by least squares; the symmetric flip channel (V3, with flip rate ε chosen on training labels) is the special case $r_0 = g_1 = \varepsilon$, $r_1 = g_0 = 1 - \varepsilon$. Proposition 1 covers this channel, since its proof uses only $P(F_t \mid w_t)$; the channel scale is chosen on training labels because reactions alone do not identify it (Proposition 2). For L_U the label event is restricted to commitments on attributes that depend on a hidden field anywhere in the schema, a set the detector can know without the clinic's rules. V6 is therefore the posterior of a proxy event (L_A of Section 4 with A the schema-level set), not of L_U: on the development corpora the two

events differ in most sessions and the labels disagree on a fifth to a quarter of them. Our simulator also violates (A1): whether a commitment is wrong depends on the intake answers, which the cell omits, so the plug-in is misspecified even with exact priors. *Prior-only* (V6−prior) keeps the fitted priors and ignores the test user's reactions; V6 minus V6−prior is the feedback increment.

Intake-aware posterior (V7). V7 repairs both mismatches and nothing else. Its cells add the intake answers each attribute depends on in the schema, shrunk toward V6's cell, which restores (A1) in our simulator. Its label event is learned: a commitment whose truth the intake fixes is wrong for every user with that intake or for none, a rule error rather than a user-specific one. Starting from the schema-level set, V7 removes an (attribute, intake) cell from the event, or returns it, when that lowers the training log-loss, alternating with the channel fit; for L, whose event is every confident commitment, only the cells change. The inputs are the agent's own intake answers and the schema's dependency lists, as for V6. V7 was designed and tuned on the development corpora only, and frozen and registered (Amendment 6) before any use on confirmatory data.

Supervised PGE classifiers. *ctx-LR* is an L_2 logistic regression ($C = 0.1$, fixed) on counts of confident commitments per (attribute, value) cell, $|H|$, the number of turns and mean confidence, with no reactions and no samples; *ctx-LR+F* adds counts per (attribute, value, reaction) cell. Neither sees the intake answers; "context-only" in the registered wording of H3 means no reactions and no samples. Both need the structured commitments that PGE elicits. Two feedback statistics (fraction of contradicted confident commitments, fraction of contradicted turns) are the simplest PGE detectors.

Baselines. SelfCheckGPT and semantic entropy on $k = 3$ resamples per turn at temperature 0.9 (exact-match clustering, adequate for categorical answers; for L_U computed on turns whose attribute is in the schema-level set above); Chain-of-Verification with the agent as its own verifier (Dhuliawala et al., 2024); mean stated (verbalized) confidence, sign chosen on the training split. Structure-based detectors from earlier work on reasoning graphs also read the reactions: a logistic regression on 14 summary features (confidence, reaction and degree statistics) of a graph with one node per commitment and edges for attribute dependencies; a two-layer GCN (Kipf and Welling, 2017); and five block-model graphon likelihoods (Lovász, 2012), of which we report the best on test, favouring the baseline.

6 Experimental Design

Environment. A simulated imaging clinic. Five coarse intake answers are shown to the agent; four hidden fields (days since treatment, allergy severity, kidney function, anxiety) are not. Each session asks for 6–12 of 12 attributes in dependency order. Six attributes can depend on a hidden field, but only for some intakes (a seventh, `transport_assistance`, depends on one in the schema but is always fixed by the intake). Ground truth is a deterministic function of intake and hidden fields; the hidden fields are independent draws from a fixed prior. The agent is a real LLM; only the patient is simulated. A second domain, a travel-booking assistant, has the same structure with different content: five intake answers (trip type, budget, travellers, duration, booking window), four undisclosed traveller facts (loyalty tier, mobility needs, dietary needs, seat preference), and twelve booking attributes, six of which can depend on an undisclosed fact. It was specified and registered before any of its data existed.

Reactions. In the main corpora the reaction to a commitment is its correctness bit flipped with probability $\varepsilon = 0.25$, absent for two attributes the patient never reacts to. In the *LLM-patient* corpus a second model (Llama-3.1-8B) role-plays the patient: it is given its own history but not the clinic's decision, one of three personas (attentive, trusting, anxious), and the agent's commitment as a hedged plain-language sentence, and replies in one or two sentences. A labeller model (in the first version, another call to the same 8B model) that sees only the sentence and the reply labels it as agreement, contradiction or neither. The agent reads the patient's verbatim replies. An *informed* variant also tells the patient the clinic's decision. A registered gate guards this corpus: a 100-session pilot runs first, and the full 600 sessions run only if contradictions are at least 1.5 times as frequent after wrong commitments as after correct ones.

Development and confirmation. All method development used two development corpora (Llama-3.3-70B, 600 sessions; Llama-3.1-8B, 200). A blind internal audit of these results found, among other issues, that the agent had been shown a corrupted feedback display (no reaction appeared as acceptance, and below $p = 0.5$ acceptance and contradiction appeared swapped), that resamples covered only five attributes, and that a feedback-free classifier nearly matched the posterior on L. Before generating new data, we committed an analysis plan to the code repository (Nosek et al., 2018): labels, detectors (V6 primary), every comparison, the statistics, and the wording each outcome licenses. Six amendments followed. The first three were committed before any confirmatory data was seen by the analysts; they added the supervised feedback increment and the label-budget curves (descriptive), the analyses of Section 7.3, and narrowed the wording of H5. The fourth added the second domain before any of its data existed; the fifth replaced the simulated patient after its first version failed its gate, before any data of the replacement existed. The sixth registered V7 and its hypotheses (H7: V7 beats V6; H8: V7 beats the supervised detector given the same information) after the registered results for four agents were known, before V7 touched any

confirmatory corpus and before the Llama-70B and travel corpora were seen. The confirmatory corpora use a fresh random seed (no session seen in development), show the agent the raw reaction, draw three resamples on every turn from that turn's own context, and cover five agents: Llama-3.3-70B, Llama-3.1-8B, Qwen2.5-72B, GPT-4o-mini and Mistral-Small-3.1-24B, 600 sessions each. Every result below is confirmatory unless marked as development.

Statistics. Splits are 70/30 by session for three split seeds; detectors are fit on the training split and score the test split once; AUROC is averaged over seeds. Differences use a paired bootstrap (2,000 resamples) clustered by session, since test sets of different seeds overlap. Pooled estimates average per-agent differences, with sessions resampled within agent. A difference is called significant when its 95% interval excludes zero; we do not correct for multiplicity. The primary hypotheses are in Table 3, the per-agent comparisons on both labels in Appendix A, and every other registered comparison in the repository's analysis output.

Oracle-information analyses. Because the simulator's rules and hidden-field prior are known, we can compute three exact quantities. The *population risk* $\bar{\eta} = \sum_h P(h)\, L_U(C,h)$ is the probability that the session's commitments would be wrong for a random user with the same intake. $\eta_\varnothing = P(L_U = 1 \mid X, C)$ adds the reactions the agent was shown (all but the last); by Theorem 1 it bounds every detector computed from the agent's context and its resamples. η_F adds the last reaction and bounds every detector of commitments and reactions. The agent's policy does not involve the hidden facts, so its factors cancel and no agent model is needed. These are not deployable detectors; they locate the information. We also give V6 and ctx-LR(+F) the same oracle knowledge of which commitments are user-dependent in each session (V6-session, LR-session). (The registered plan calls $\bar{\eta}$, $\eta_\varnothing$ and η_F eta0, etaX and etaF.)

7 Results

7.1 Sampling cannot see the user

Table 1 shows the mechanism. On commitments whose truth depends on undisclosed facts, agents that are self-consistent repeat the same answer whether it is right or wrong (e_0 and e_1 small and close, so by Corollary 1 the AUROC of disagreement lies within $\frac{1}{2}e_0e_1$ of $\frac{1}{2} + \frac{1}{2}(e_1 - e_0)$), so disagreement carries little information. One reaction from our simulated user gives 0.75, as designed; the comparison shows how little a user's reaction needs to carry to beat resampling, not how much a real user's does. At the session level (Table 2), SelfCheckGPT and semantic entropy score 0.566 [0.535, 0.595] and 0.494 [0.479, 0.510] pooled over agents, both intervals inside the registered band of $[0.40, 0.60]$ (H1): sampling detectors are near chance on user-specific hallucination. H1 is a pooled claim: per agent SelfCheckGPT ranges over 0.520–0.610; the top value (Llama-8B) lies just outside the band. They are not near chance everywhere. On the composite label, where many errors are guesses about clinic rules that do vary across resamples, SelfCheckGPT reaches 0.676 [0.645, 0.707], and on AgentBench database questions (Liu et al., 2024), where errors are computational, it reaches 0.712 with the 70B agent (a development corpus from before the audit, described in the repository). The failure is specific to facts the user never disclosed.

Table 1: Per-commitment check of Corollary 1 on commitments (any confidence) whose truth is user-dependent in that session. e_1, e_0: probability that some resample differs from the commitment, given that it is wrong / right. AUROC for predicting *that* commitment's error; the reaction column is on commitments that received one and equals $1 - \varepsilon = 0.75$ by the simulator's design.

Agent	wrong	all agree	e_1	e_0	Disagreement	One reaction
Llama-70B	??%	??%	??	??	??	??
Llama-8B	39.7%	49.5%	0.544	0.478	0.555	0.746
Qwen-72B	43.1%	83.1%	0.189	0.154	0.517	0.744
GPT-4o-mini	42.9%	85.6%	0.183	0.115	0.535	0.746
Mistral-24B	40.6%	69.2%	0.344	0.283	0.534	0.746

7.2 Detecting user-specific hallucination

Table 2 gives the confirmatory results and Table 3 the registered hypotheses, pooled over four agents.

Registered detectors. The user's implicit reactions add detectable signal beyond population base rates: +.072 [+.054, +.090]* to the posterior over its cell priors (H2; significant for 4 of 4 agents), and beyond the commitments themselves for the supervised detector (+.058 [+.038, +.080]*, H2b). Detectors on PGE observables beat both sampling detectors (H4: +.090 [+.056, +.123]* and +.162 [+.127, +.195]*; H4b: +.139 [+.101, +.176]* and +.211 [+.179, +.242]*). Against the registered supervised detectors the first posterior, V6, did not win: the comparison with the commitment-only classifier is null (H3: +.008 [−.028, +.045]; registered wording: "context alone matches PGE on this label"), and ctx-LR+F is ahead with the full training split (−.050 [−.084, −.019]*). V3, with a symmetric channel, is indistinguishable from V6 (−.002 [−.007, +.002]).

The intake-aware posterior (Amendment 6). These results, and the oracle analysis of Section 7.3, located V6's weaknesses: its cells omit the intake, and its event is a proxy for the commitments the intake leaves to the user. V7 repairs both (Section 5). It reaches 0.806 [0.784, 0.828] AUROC pooled (Table 2). Conditioning on the intake improves the posterior (H7: +.150 [+.123, +.180]*), and given the same information (intake, commitments, reactions) the posterior beats the supervised detector (H8:

Table 2: Session AUROC on the user-specific label L_U (mean of three splits; confirmatory corpora, 600 sessions per agent). †Exploratory, not registered: ctx-LR+F plus the intake answers and commitment×intake cells. ‡Registered in Amendment 6 after the other results were known; designed on development data only. CoVe was run for the 70B agent only. The composite label L is in the repository.

	Detector	Llama-70B	Llama-8B	Qwen-72B	GPT-4o-mini	Mistral-24B
Baselines	SelfCheckGPT	??	.610	.559	.576	.520
	Semantic entropy	??	.533	.485	.486	.474
	CoVe	??	–	–	–	–
PGE: commitments	Mean confidence	??	.512	.522	.465	.454
	ctx-LR	??	.686	.715	.635	.556
	V6−prior	??	.591	.593	.575	.577
PGE: commitments + reactions	Contradicted confident	??	.614	.628	.576	.637
	Contradicted turns	??	.536	.579	.595	.624
	Graph-feature LR	??	.647	.653	.609	.596
	GCN	??	.574	.663	.597	.569
	Best graphon	??	.531	.506	.521	.409
	ctx-LR+F	??	.725	.748	.668	.682
	Intake-LR+F†	??	.782	.749	.717	.690
	PGE posterior (V6)	??	.672	.650	.636	.666
	PGE posterior, intake-aware (V7)‡	??	**.889**	**.810**	**.836**	**.690**
	L_U positive rate	??%	29.7%	39.0%	35.3%	40.3%

+.071 [+.047, +.101]*; significant for 3 of 4 agents, the remaining one a tie). It also beats the registered supervised detector (+.100 [+.074, +.129]*) and both sampling detectors by +.240 [+.206, +.276]* and +.312 [+.284, +.339]* (significant for 4 of 4 and 4 of 4 agents). On the composite label it reaches 0.926 [0.910, 0.938] and again beats the supervised detector with the same information (+.023 [+.004, +.039]*).

Label budgets. Labels for user-specific hallucination require knowing the user's private facts, so small budgets are the realistic case (Figure 1; descriptive). With 25 labelled sessions V7 leads both supervised detectors on 4 of 4 agents (mean 0.632, against 0.603 for intake-LR+F and 0.583 for ctx-LR+F); the two tie on average at 50, and V7 is ahead again at 100 and 200 (0.757 against 0.715).

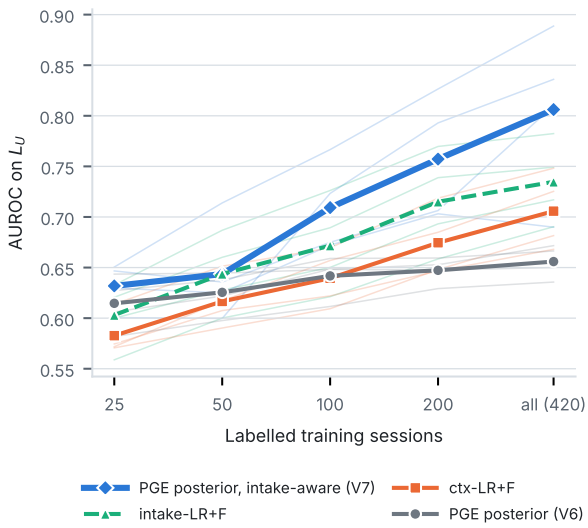


Figure 1: Test AUROC on L_U against the number of labelled training sessions (mean over agents; thin lines: individual agents; identical training subsets for every detector).

7.3 Where the information is

Table 4 and Figure 2 locate the signal. The population risk $\bar{\eta}$ alone is far above every sampling detector: much of L_U is visible in *which* values an agent commits to, given the intake. The user's reactions add $\eta_F - \bar{\eta}$ beyond it (H5). Almost all of that information is already in the conversation the agent sees ($\eta_F - \eta_\varnothing$ is +.001 [−.001, +.004]), so the failure of sampling detectors is not that the information is absent from their input; it is that agreement between resamples does not read it. Given the same information (which commitments concern undisclosed facts), the PGE posterior beats the supervised detector (H6), though by a small margin whose interval nearly reaches zero (+.017 [+.0004, +.036]*). Registered without a claim: V6-session − $\bar{\eta}$ is +.033 [+.010, +.056]*, and the reactions add +.034 to +.148 to LR-session across agents. Without that knowledge, V6 targets a schema-level set of attributes that is only a proxy for the label's support, which costs it 0.181–0.233 AUROC against V6-session, an order of magnitude more than the H6 margin: knowing *which* commitments concern undisclosed facts matters far more than how the evidence is combined. A deployable detector has to

Table 3: Registered hypotheses on L_U, pooled over four agents (stratified session bootstrap; *: 95% CI excludes 0). H5 and H6 use oracle information (Section 7.3); H7 and H8 are Amendment 6.

	Comparison	Estimate [95% CI]
H1	AUROC(SelfCheckGPT)	0.566 [0.535, 0.595]
H1	AUROC(semantic entropy)	0.494 [0.479, 0.510]
H2	V6 − V6−prior (feedback increment)	+.072 [+.054, +.090]*
H2b	ctx-LR+F − ctx-LR (feedback increment)	+.058 [+.038, +.080]*
H3	V6 − ctx-LR	+.008 [−.028, +.045]
H4	V6 − SelfCheckGPT	+.090 [+.056, +.123]*
H4	V6 − semantic entropy	+.162 [+.127, +.195]*
H4b	ctx-LR+F − SelfCheckGPT	+.139 [+.101, +.176]*
H4b	ctx-LR+F − semantic entropy	+.211 [+.179, +.242]*
–	V6 − ctx-LR+F	−.050 [−.084, −.019]*
H5	$\eta_F - \bar{\eta}$	+.087 [+.069, +.107]*
H6	V6-session − LR-session+F	+.017 [+.0004, +.036]*
H7	V7 − V6	+.150 [+.123, +.180]*
H8	V7 − intake-LR+F (same information)	+.071 [+.047, +.101]*
H8′	V7 − ctx-LR+F	+.100 [+.074, +.129]*

Table 4: Oracle-information analysis on L_U (AUROC). $\bar{\eta}$: population risk of the commitments; $\eta_\varnothing$: Bayes given the conversation the agent saw (Theorem 1's bound for context-and-sample detectors); η_F: Bayes given all reactions. V6-session and LR-session+F are told which commitments are user-dependent in each session. Pooled differences are in Table 3.

Agent	$\bar{\eta}$	$\eta_\varnothing$	η_F	V6-sess.	LR-sess.+F
Llama-70B	??	??	??	??	??
Llama-8B	.899	.952	.953	.904	.908
Qwen-72B	.854	.930	.932	.879	.851
GPT-4o-mini	.803	.907	.909	.837	.825
Mistral-24B	.781	.890	.891	.847	.814

learn this from labels. A generic classifier mostly does not (intake-LR+F reaches 0.690–0.782, against 0.781–0.899 for the population risk computed with the rules). V7, which learns it as the posterior's label event, closes 71% of V6's gap to V6-session (mean 0.867) without oracle knowledge.

7.4 A second domain

[PENDING: second-domain paragraph: does the pattern transfer (sampling near chance, population risk large, reactions add, V6 vs supervised)?]

7.5 Natural-language reactions

The first simulated patient failed its gate. It contradicted 83% of wrong commitments and 79% of correct ones (lift 1.06; 100 sessions, 817 replies), against 75% and 25% for the noisy bit; the informed variant, told the clinic's decision, did little better (97% and 80%, lift 1.21). Reading the replies shows why: the patient voiced doubt about facts it had not been given, and the labeller counted doubt as contradiction. As registered, neither was run at scale. A

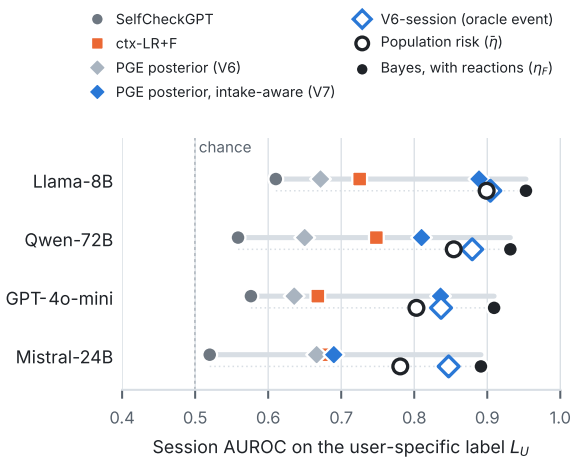


Figure 2: Where the signal is, on L_U. Upper row of each agent: deployable detectors. Sampling sits near chance; the intake-aware posterior V7 is the best of them. Lower row: oracle references (V6 told the user-dependent set, the population risk $\bar{\eta}$ and the Bayes posterior with reactions η_F).

Table 5: Second domain (travel booking), AUROC on L_U, 600 sessions per agent.

Agent	SelfCheck	ctx-LR	ctx-LR+F	V6	$\bar{\eta}$	η_F
GPT-4o-mini	??	??	??	??	??	??
Llama-8B	??	??	??	??	??	??

second version, registered before its data existed (Amendment 5), uses GPT-4o-mini for both roles, lets the patient object only to commitments that conflict with its listed facts, and labels a reply as a contradiction only if it states a conflict. [PENDING: v2 audit (P(-1|wrong), P(-1|right), lift) and, if it passes, detection V6 / V6-prior / ctx-LR / ctx-LR+F on both labels] A simulated user that does not track its own facts produces nearly uninformative reactions; the gate is what keeps such a corpus out of the results.

7.6 Choosing what to ask

The posterior's per-commitment risk q_t is what chooses the question, and no session-level classifier provides it. On confident user-dependent commitments in the test splits, V7's q_t predicts that commitment's error with AUROC 0.765–0.803 and expected calibration error 0.049–0.077 across agents. On the same commitments the reaction alone gives 0.740–0.761 ($1 - \varepsilon$ by design), resample disagreement 0.520–0.577 and stated confidence 0.484–0.552 (descriptive; q_t includes the commitment's own reaction and adds the population risk of its cell). The measured loop (70B agent) uses the first 120 test sessions of split seed 42 instead of the registered 180, a deviation forced by the API budget. [PENDING: measured clarification: posterior@1 vs no-question, random, lowest-confidence; pos-

terior@2 vs detailed questions and the full-rerun control; questions per session; primary outcome wrong confident user-dependent commitments]

7.7 Robustness to the user's behaviour

Our users react with a designed noise level. To see how much that matters, we regenerated the reactions seen by the detectors on the confirmatory trajectories, with the agents' trajectories fixed (Theorem 2's setting; exploratory). As the flip rate rises from 0.05 to 0.45, V7's mean AUROC on L_U falls from 0.957 to 0.683, tracking the Bayes bound (0.988 to 0.840); it leads intake-LR+F at every flip rate up to 0.35 and ties it at 0.45. When only half of the commitments receive a reaction it still leads (0.748 vs. 0.712); at a quarter it trails (0.666 vs. 0.688), failing on one agent. SelfCheckGPT stays at 0.566 throughout.

On the development corpora (protocol v2; label L) we also replaced the reactions seen by the detectors, not the agent, with channels the posterior does not assume: symmetric and asymmetric flips, attribute-dependent noise, extra missingness, and a "lazy" user who contradicts 60% of wrong commitments and never confirms correct ones. The symmetric posterior V3 collapses under the lazy user (0.425 for 70B, 0.576 for 8B), reading the absence of confirmations as evidence of correctness; the general channel of V6 recovers it from data (0.857 and 0.855). Across all ten channels V6 has the best point estimate on both agents among the then-implemented detectors (significant in six channels for 70B and two for 8B). Structure does not help: graphon likelihoods score 0.43–0.58 on every development corpus and label, and exact inference in a cascade model over the dependency graph is no better than independence (0.732 vs. 0.734).

8 Limitations

Simulation. The agents are real, but the patient is simulated: by a noisy correctness bit in the main corpora and by an LLM in one corpus. The oracle analyses rely on the simulator's known rules and prior, which a deployment does not have. *Two synthetic domains.* Both private-attribute domains are simulators we designed, with deterministic rules and independent hidden facts; real user populations have correlated, partially self-reported facts. *Post-hoc method.* V7 was designed after the registered results were known, on development data only, and registered before use; Llama-70B and the travel domain are its fully unseen tests. *Labels.* Labels are programmatic and noise-free by construction; there is no human evaluation. *Supervision.* The posterior and the supervised baselines are fit on labelled training sessions; the posterior can run without labels when the channel is known, which we only test in development. *Untested detectors.* A second model that reads the transcript with its reactions is a feedback detector, not bounded by Theorem 1; we did not test one. Sampling detectors used three resamples (ten were registered as optional and not run), and with exact matching semantic entropy takes few values. We did not run hidden-state probes; Theorem 1 predicts they share the within-context limit, but they could exploit population risk as ctx-LR does. *Theory.* Proposition 1 is exact only under (A1)–(A2) with the true priors and channel, and the detector is a plug-in estimate; Theorem 2 covers reactions regenerated for the detector with the agent's trajectories fixed, not the protocol's own noise, because the agent conditions on the reactions it is shown.

9 Conclusion

When an agent invents a fact about its user, its own samples cannot tell which of the users sharing its context it is wrong about. Some of the risk is visible in what the agent commits to; the rest is visible only through the user's reactions. Making the agent state its assumptions as probabilistic commitments and recording the user's reactions makes both available, and a Bayes analysis in a simulator with known truth shows how much each is worth. An intake-aware posterior turns them into the best detector we tested, one that also says which commitment to question. [PENDING: one sentence on measured clarification]

Reproducibility. Code, all trajectories, the pre-registered plan with its five amendments and a log of their commit times, every analysis script, and a ledger of all runs including discarded variants are at [PENDING: anonymous repository link].

Use of AI tools. [PENDING: AI-assistance disclosure per ICAART policy, without names.]

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A Per-agent registered comparisons

Table 6 gives every registered difference per agent on both labels (paired session-clustered bootstrap; *: 95% CI excludes 0). H2: V6 − V6−prior; H2b: ctx-LR+F − ctx-LR; H3: V6 − ctx-LR; H4: V6 − SelfCheckGPT (SC) and semantic entropy (SE).

Table 6: Registered differences per agent (AUROC).

Agent	H2	H2b	H3	H4 (SC)	H4 (SE)	V6−ctx-LR+F
User-specific label L_U (primary)						
Llama-70B	??	??	??	??	??	??
Llama-8B	+.079 [+.042, +.116]*	+.040 [+.002, +.079]*	−.016 [−.082, +.052]	+.060 [−.015, +.134]	+.139 [+.067, +.209]*	−.056 [−.111, −.001]*
Qwen-72B	+.057 [+.023, +.091]*	+.033 [−.009, +.075]	−.070 [−.137, −.002]*	+.086 [+.014, +.162]*	+.159 [+.096, +.220]*	−.103 [−.163, −.044]*
GPT-4o-mini	+.060 [+.026, +.095]*	+.034 [−.004, +.075]	+.005 [−.070, +.079]	+.058 [+.004, +.111]*	+.151 [+.090, +.215]*	−.029 [−.099, +.040]
Mistral-24B	+.090 [+.052, +.127]*	+.126 [+.077, +.175]*	+.115 [+.042, +.187]*	+.150 [+.079, +.222]*	+.197 [+.138, +.255]*	−.011 [−.073, +.051]
Pooled	+.072 [+.054, +.090]*	+.058 [+.038, +.080]*	+.008 [−.028, +.045]	+.090 [+.056, +.123]*	+.162 [+.127, +.195]*	−.050 [−.084, −.019]*
Composite label L (secondary)						
Llama-70B	??	??	??	??	??	??
Llama-8B	+.113 [+.074, +.152]*	+.067 [+.037, +.099]*	+.086 [+.028, +.145]*	+.186 [+.115, +.257]*	+.255 [+.187, +.322]*	+.018 [−.022, +.058]
Qwen-72B	+.085 [+.054, +.117]*	+.028 [+.004, +.054]*	+.031 [−.002, +.067]	+.250 [+.181, +.320]*	+.352 [+.301, +.399]*	+.003 [−.026, +.033]
GPT-4o-mini	+.037 [+.014, +.062]*	+.005 [−.025, +.036]	+.042 [−.004, +.092]	+.111 [+.048, +.176]*	+.269 [+.203, +.331]*	+.037 [−.004, +.082]
Mistral-24B	+.087 [+.055, +.121]*	+.026 [−.007, +.059]	−.005 [−.049, +.042]	+.135 [+.058, +.209]*	+.315 [+.248, +.378]*	−.030 [−.072, +.012]
Pooled	+.080 [+.064, +.096]*	+.031 [+.017, +.047]*	+.040 [+.015, +.063]*	+.175 [+.135, +.207]*	+.302 [+.269, +.329]*	+.009 [−.012, +.027]